

Data Science & AI Foundations

You've been surrounded by data science your whole life. M-Pesa runs on it. Safaricom's network runs on it. The crop forecast your uncle heard on the radio — that too. This week, we name the pieces.



Duration: 1 week

Hands-on: Iris dataset + Zindi Challenge

Assessment: Quiz + Discussion

By end of this week, you will be able to

Define Data Science, Machine Learning, and Artificial Intelligence — and tell them apart

Describe the Data Science Lifecycle from raw data to deployed model

Identify real AI applications in Kenyan and African contexts

Explain why bias and fairness matter in AI systems

Load and explore a real dataset in Python

1. What is Data Science?

Start with a question: *How does M-Pesa decide to flag a suspicious transaction in under two seconds?*

There is no human manually reviewing your Ksh 50 transfer. A model — trained on millions of past transactions — makes that call. It learned patterns: typical transaction sizes, times, locations, recipient relationships. When your transfer breaks those patterns, it raises a flag. That process — from raw transaction logs to an automated decision — is data science.

Data science sits at the intersection of three things: **statistics** (understanding data), **programming** (working with data at scale), and **domain knowledge** (knowing what the data means in context). You already have the domain knowledge. This program gives you the other two.

Quick distinction: Data Science is the full process — from collection to insight to deployment. Machine Learning is one tool within that process (building models that learn from data). Artificial Intelligence is the broader goal — making machines that can reason, recognise, and decide.

AI in your world right now



Healthcare

AI screening chest X-rays for TB at KEMRI. Chatbots triaging patients at Jubilee Insurance.



Finance

M-Shwari credit scoring. Fraud detection at Equity Bank. Real-time FX models at NSE.



Agriculture

PlantVillage crop disease detection. Satellite yield prediction by Apollo Agriculture.



Transport

Bolt and Uber surge pricing. Swvl route optimisation. NTSA accident risk modelling.

2. The Data Science Lifecycle

Every data science project — from predicting matatu delays to diagnosing malaria from blood smears — follows the same six-stage loop.

Define

What problem are we solving? What does success look like?

Collect

Surveys, sensors, APIs, scrapers, hospital records

Clean

Handle missing values, fix formats, remove duplicates

Explore

Visualise, find patterns, spot outliers

Model

Train, evaluate, and compare algorithms

Deploy

Put the model to work. Monitor it. Update it.

Loop, not line: You will almost always go backwards — realising in the modelling stage that your data collection missed something, or that the problem definition needs refining. That is normal. That is the job.

3. Supervised, Unsupervised & Deep Learning

Type	What it does	Kenyan example
Supervised Learning	Learns from labelled data — input + correct answer pairs	M-Shwari loan default prediction (historical repayment data)
Unsupervised Learning	Finds patterns in unlabelled data on its own	Grouping Safaricom users by behaviour for targeted offers

Deep Learning

Uses many-layered neural networks, great for images and speech

Recognising maize leaf disease from smartphone photos

4. Responsible AI: When Algorithms Harm

AI systems learn from historical data. If that history contains human bias, the model learns the bias too — and then scales it to millions of decisions.

⚠ Case Study: Facial Recognition and Dark Skin

A 2018 MIT study (Joy Buolamwini) found that major commercial facial recognition systems misidentified dark-skinned women 35% of the time, versus 1% for light-skinned men. The models were trained mostly on images of light-skinned faces. In 2020, Detroit police arrested an innocent Black man based on a facial recognition match. He spent days in custody before the error surfaced.

Why this matters in Africa: As Kenyan institutions adopt AI for identity verification, credit scoring, and policing, biased training data produces outcomes that systematically disadvantage specific communities. Knowing how bias enters a model — and how to detect it — is not optional knowledge for you as a data scientist.

Three types of bias to know

- **Historical bias:** the real world contained unfair outcomes, your data captured them
- **Representation bias:** your training data under-represents certain groups
- **Measurement bias:** the way you collected or labelled data was not consistent across groups

5. Hands-On: Exploring the Iris Dataset

The Iris dataset is the "Hello, World" of data science. It contains 150 measurements of flower petals and sepals from three iris species. Small enough to understand completely. Interesting enough to reveal real patterns.

Open **Google Colab** (colab.research.google.com) — no installation needed. Create a new notebook and run the following code blocks one by one.

Step 1: Load and inspect the data

```
# Import our tools import pandas as pd import matplotlib.pyplot as plt from
sklearn.datasets import load_iris # Load the dataset iris = load_iris() df =
pd.DataFrame(iris.data, columns=iris.feature_names) df['species'] =
pd.Categorical.from_codes(iris.target, iris.target_names) # First look:
shape and types print(f"Dataset shape: {df.shape}") print("\nFirst 5 rows:")
df.head()
```

Step 2: Summary statistics

```
# What does the data look like numerically? df.describe()
```

Step 3: Visualise the distributions

```
# Plot petal length by species fig, ax = plt.subplots(figsize=(8, 5)) for
species, group in df.groupby('species'): ax.scatter(group['petal length
(cm)'], group['petal width (cm)'], label=species, alpha=0.7)
ax.set_xlabel('Petal Length (cm)') ax.set_ylabel('Petal Width (cm)')
ax.set_title('Iris: Petal Dimensions by Species') ax.legend()
plt.tight_layout() plt.show()
```

Step 4: Your turn — investigate

```
# Try these yourself: # 1. Which species has the longest sepals on average?
df.groupby('species')['sepal length (cm)'].mean() # 2. Are there any missing
values? df.isnull().sum() # 3. Plot sepal length vs sepal width # (copy the
scatter plot above and change the columns)
```

Observation to make: After running Step 3, you will notice one species separates cleanly from the other two. Write down which one, and why you think the petal dimensions alone are enough to identify it.

Resources & References

Curated readings, courses, datasets, and tools for Week 1. Free resources are marked — no excuses.

Free Courses & Interactive Learning

[Freekaggle.com](#) ↗

Intro to Machine Learning — Kaggle Learn

Decision trees, model validation, underfitting vs overfitting. Browser-based, no setup. Best first ML course.

[Freegoogle.com](#) ↗

ML Crash Course — Google

TensorFlow-based intro from Google engineers. Covers linear regression, loss functions, and gradient descent with interactive exercises.

[Freefast.ai](#) ↗

Practical Deep Learning for Coders — fast.ai

Top-down approach: start with a working model, then understand it. Used by practitioners worldwide.

[Audit Freecoursera.org](#) ↗

Machine Learning Specialization — Andrew Ng

The gold-standard ML introduction. 3-course series covering supervised and unsupervised learning. Audit at no cost.

[Free via Ngdatacamp.com](#) ↗

Understanding Machine Learning — DataCamp

Conceptual ML intro, no coding required. Your DataCamp Donates access covers this. Perfect week 1 companion.

[Freeelementsofai.com](#) ↗

Elements of AI — University of Helsinki

Plain-English AI concepts with no maths prerequisites. Great for building mental models before diving into code.

Books & Reading

Free PDF statlearning.com ↗

An Introduction to Statistical Learning (ISLR)

James, Witten, Hastie & Tibshirani. The foundational applied ML textbook — concepts explained with minimal jargon. PDF free on the official site.

Free Online [github.io](https://github.com) ↗

Python Data Science Handbook — Jake VanderPlas

Covers NumPy, Pandas, Matplotlib, and Scikit-learn in depth. The practical companion to everything in this curriculum.

Free Notebooks github.com ↗

Hands-On Machine Learning — Aurélien Géron

Scikit-learn + TensorFlow. Industry favourite. All Jupyter notebooks are free on GitHub even without the book.

Africa-Focused Resources

Africa zindi.africa ↗

Zindi Africa

Africa's largest data science competition platform. Real problems from African organisations. Explore beyond the assigned challenges.

Africa datascienceafrica.org ↗

Data Science Africa

Annual summer school across Kenya, Uganda, Tanzania, Rwanda, and Nigeria. Published proceedings and lecture videos freely available.

Africa masakhane.io ↗

Masakhane — NLP for African Languages

Grassroots research community building NLP models for Swahili, Amharic, Yoruba, and 50+ more African languages.

Africa indabax.co.za ↗

Deep Learning IndabaX Kenya

Pan-African deep learning summer school with a Kenyan chapter. Workshop materials and lecture recordings are publicly available.

Datasets to Explore

Free uci.edu ↗

UCI Machine Learning Repository

650+ clean datasets for practice. The Iris dataset lives here. Good for Week 1 exploration and model comparison exercises.

Free kaggle.com ↗

Kaggle Datasets

Hundreds of thousands of datasets. Filter by "Kenya" or "Africa" to find locally relevant data for your projects.

Kenya Gov opendata.go.ke ↗

Kenya Open Data Portal

Official government datasets: census, health, agriculture, infrastructure. Ideal for capstone projects with real Kenyan context.

Academic References

- [1] Buolamwini, J. & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. *Proceedings of Machine Learning Research*, 81, 1–15. [proceedings.mlr.press ↗](https://proceedings.mlr.press/v81/buolamwini18.html)
- [2] James, G., Witten, D., Hastie, T. & Tibshirani, R. (2023). *An Introduction to Statistical Learning with Applications in Python*. Springer. [Free PDF ↗](#)
- [3] VanderPlas, J. (2016). *Python Data Science Handbook*. O'Reilly Media. [Free online ↗](#)
- [4] Abebe, R., et al. (2020). Roles for Computing in Social Change. *ACM FAccT 2020*. — Covers how algorithmic systems can entrench or challenge inequality.
- [5] Ndung'u, N. & Signé, L. (2020). The Fourth Industrial Revolution and Digitisation will Transform Africa into a Global Powerhouse. *Brookings Institution*. [brookings.edu ↗](https://www.brookings.edu/research/the-fourth-industrial-revolution-and-digitisation-will-transform-africa-into-a-global-powerhouse/)
- [6] Fisher, R.A. (1936). The Use of Multiple Measurements in Taxonomic Problems. *Annals of Eugenics*, 7(2), 179–188. — The original Iris dataset paper.

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Questions? Post in the Week 1 Discussion Forum or reach your mentor during the weekly check-in.